Stock Segmentation of the S&P 500: A Business Intelligence Approach

Module: 7BUIS010W – Data Warehousing and Business Intelligence
Group Coursework

GitHub Repository:

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Introduction

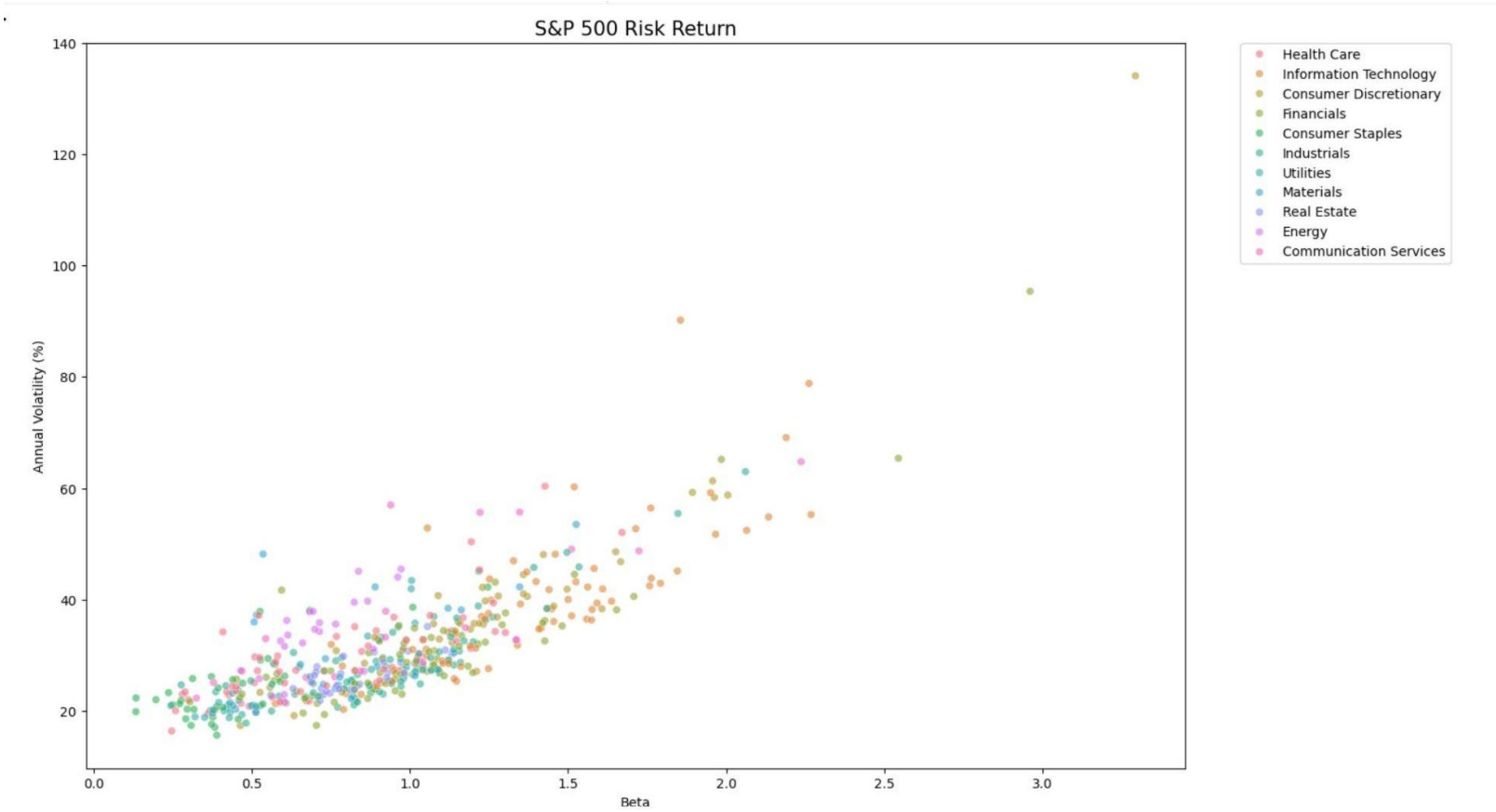
This report presents a comprehensive Business Intelligence analysis of the S&P 500 index to support data driven investment decisions. Using Python in Google Colab, we extracted, cleaned, and analysed daily market data for 496 stocks (after removing invalid entries) from 1 January 2022 to 1 January 2025 via the `yfinance` library.
Key risk-return metrics Mean Daily Return, Beta, and Annual Volatility were calculated and used for stock segmentation through agglomerative hierarchical clustering and K-Means. The analysis helps investors build diversified portfolios aligned with specific risk tolerances. Findings are translated into BI recommendations, including a proposed star schema data mart and insights on CRM systems for sustainability in Fintech. All code, data processing steps, and visualisations are available in the public GitHub repository.

Quantitative Risk Assessment and Metrics

The metrics of Mean Daily Return, Beta, and Annual Volatility serve as the pillars for quantitative risk assessment in this analysis. While Mean Daily Return indicates the average growth potential, Beta and Annual Volatility provide a nuanced view of the risk profile essential for strategic portfolio construction.

> **Beta:** Acts as the primary measure of systematic risk, revealing a stock's sensitivity to broader S&P 500 market movements. Values exceeding 1 identify aggressive assets, while those below 1 signify defensive, market resilient holdings.
>
> **Annual Volatility:** Complements this by quantifying total risk, capturing the intensity of price fluctuations irrespective of market direction.

When visualised, the relationship between these metrics reveals a risk map where the use of sector-based colouring transforms the data into identifiable business segments.

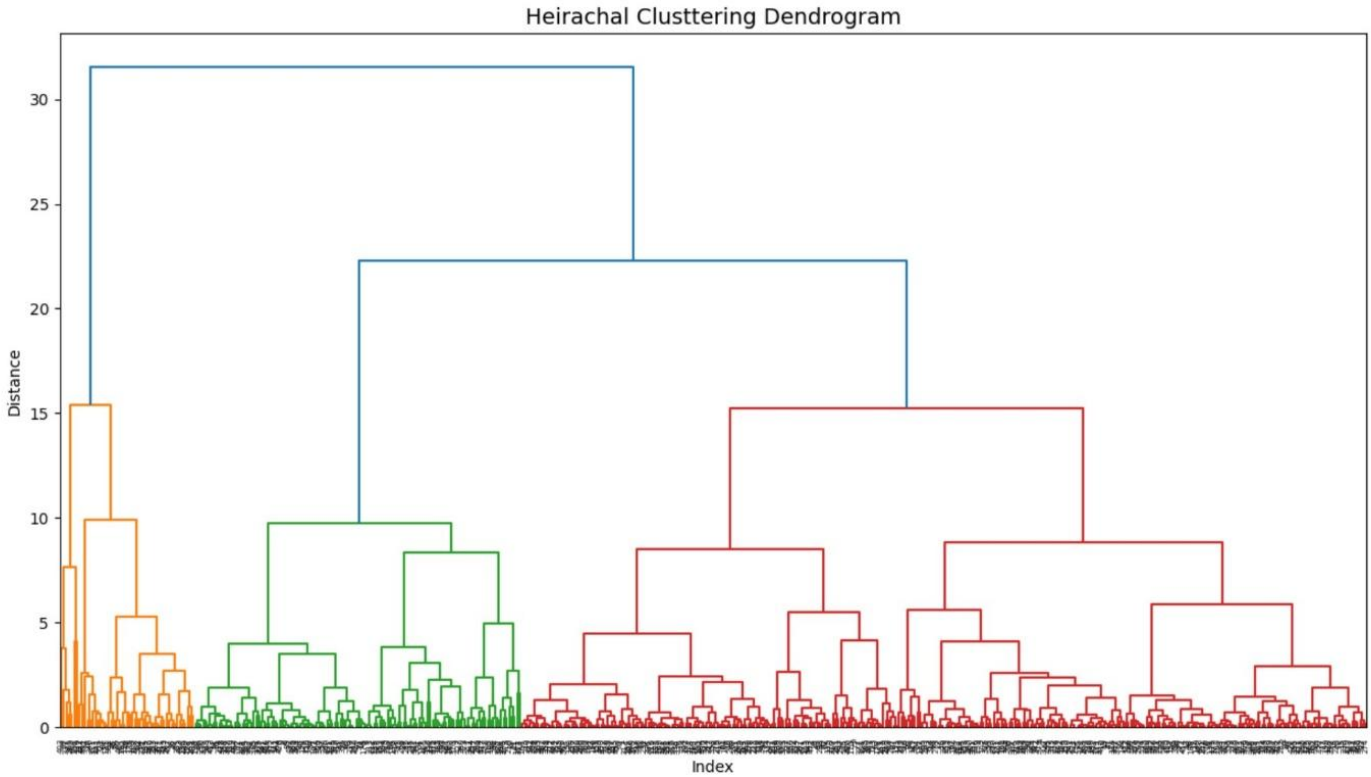


Agglomerative Hierarchical Clustering

To perform the initial segmentation of the S&P 500 Agglomerative Hierarchical Clustering was used. This bottom up approach is particularly suitable for financial data sets as it does not require a predefined number of clusters, allowing the natural hierarchy of market sectors to emerge. Using Ward's Linkage, the model minimises within cluster variance, ensuring that stocks grouped together possess highly similar risk return properties. To facilitate this 'Mean daily return, Beta and Annual volatility' were processed using the StandardScaler. This ensures that the Beta values are not ignored and treated with equal weight to volatility, despite having vastly different ranges.

Identifying the optimal number of Clusters

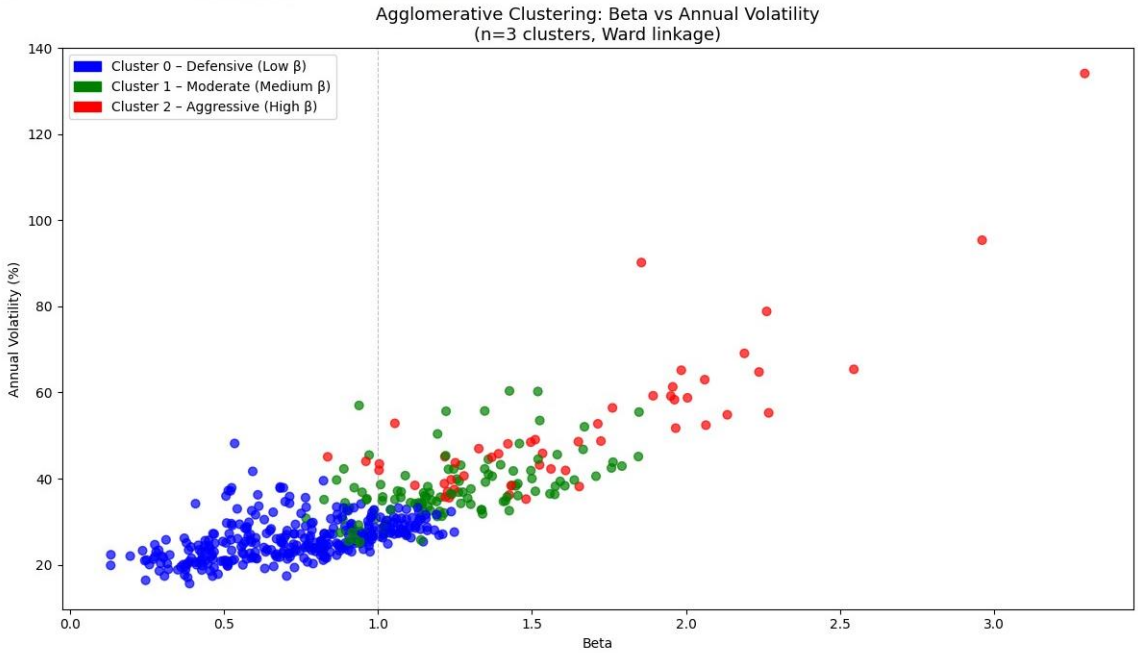
To identify the best value for N, we designed a Dendrogram, which visually represents the hierarchical relationships and distances between stock clusters. By analysing the vertical distances, we looked for the largest vertical gap that does not intersect any horizontal cluster merging lines. Drawing a horizontal cut through these longest vertical branches provides the most stable cluster count. In our analysis, cutting the dendrogram at approximately 18 yields N=3 distinct clusters, as indicated by the three primary vertical branches (Orange, Green, and Red in our visualisation). We chose this because it captures significant variance between groups, separating stocks by their unique combinations of Daily Return, Beta, and Annual Volatility without over segmenting the data into too many small, non-distinct groups.

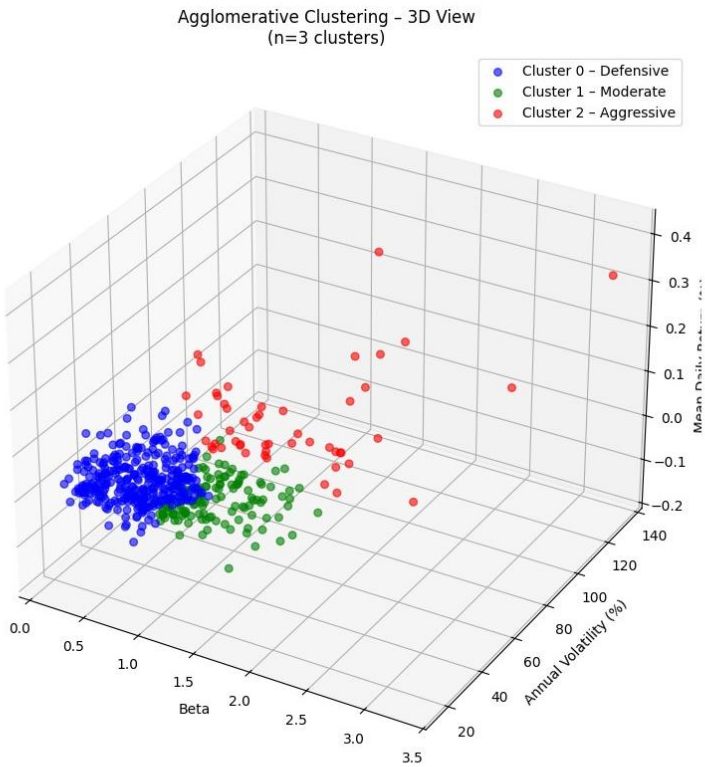


The Index (X - axis) - Individual stocks

Distance (Y - axis) - Measures how far apart two stocks are based on their mean return, beta and volatility.

Agg_Cluster	Mean_Daily_Return_Pct	Beta	Annual_Volatility_Pct
0	0.0445	0.7232	25.9383
1	-0.0023	1.2229	37.0634
2	0.1541	1.6529	52.2985






```
Cluster risk labels assigned:
Agg_Cluster_Label
Defensive (Low β)      321
Moderate (Medium β)    124
Aggressive (High β)     51
```

This scatter plot visualises the three distinct investment regimes. By plotting Beta against Annual Volatility, we can see a clear progression of risk:

- **Cluster 0 (Blue) – Defensive:** These stocks exhibit the lowest systematic risk and volatility, making them ideal for capital preservation.
- **Cluster 1 (Green) – Moderate:** These assets track the broader market closely, representing a balanced middle ground.
- **Cluster 2 (Red) – Aggressive:** These stocks surpass the Beta = 1.0 threshold and exhibit the highest volatility, identifying them as high risk, high reward assets.

K-Means Clustering and Skew Management

K-means clustering is highly appropriate for this analysis as it effectively groups stocks based on similarity across multiple numerical features: daily return, beta, and annual volatility. Since these variables are continuous and numeric, they are well suited for K-means, which utilises Euclidean distance to partition data into distinct clusters.

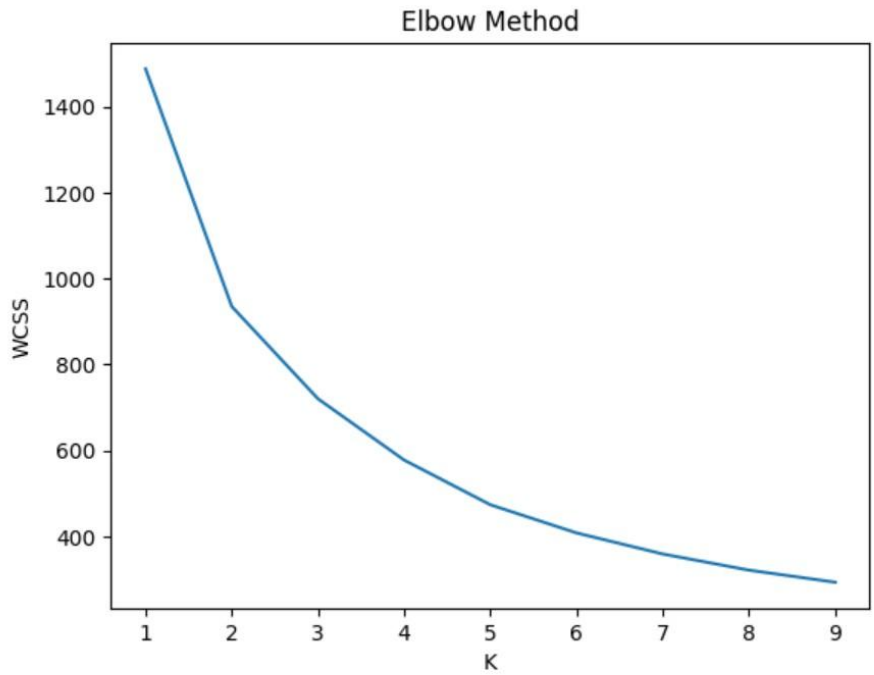
The primary objective is to identify stocks with similar risk reward profiles. K-means achieves this by iteratively minimising the distance between stocks and their respective cluster centroids. With a dataset of roughly 496 stocks, K-means is significantly more computationally efficient than hierarchical methods. It produces non overlapping groupings that are easily interpretable for investment strategists, making it an ideal tool for large scale financial segmentation.

Identifying the Optimal K

Two key techniques were used to the Elbow Method and the Silhouette Score to determine the optimal number of clusters (k) The Elbow Method works by plotting the Within-Cluster Sum of Squares (WCSS) against the different values of K. As the number of clusters increases, the WCSS decreases because the clusters become more refined and closer to the data points. However, after a certain point, the improvement starts to slow down. This point is known as the “elbow” and it suggests the most suitable number of clusters, as increasing K further does not significantly improve the model.

The Silhouette Score was also used to support this. It measures how well each stock fits within its assigned cluster compared to other clusters. A higher score means the clusters are more clearly separated and better defined. By comparing the silhouette scores for different values of K, it becomes easier to choose the most appropriate number of clusters.

Before applying K-means, the data needed to be prepared properly. Financial data such as daily returns and volatility can often be skewed and may include extreme values. To deal with this, standardisation i.e scaling was applied so that all variables are on the same scale and contribute equally to the clustering process. This is important because without scaling, one variable could dominate the results. Overall, this preprocessing step helps to improve the accuracy and reliability of the clustering results.




```
K=2, Score=0.4621945316808228
K=3, Score=0.31096782951836566
K=4, Score=0.32600659528994347
K=5, Score=0.29846659843549245
K=6, Score=0.2913806447851148
K=7, Score=0.2939849162157132
K=8, Score=0.29135890345583393
K=9, Score=0.29133532974916
         Mean_Daily_Return_Pct      Beta  Annual_Volatility_Pct
Cluster
0                     0.004759  1.170004              35.985756
1                     0.046942  0.692432              25.277772
2                     0.153019  1.705586              53.244110
```

The clustering results can be interpreted by grouping the stocks based on their level of risk using beta and annual volatility. Cluster 0 represents the low risk group, where stocks have lower beta values and lower annual volatility. This means they are more stable and do not react as strongly to changes in the market, making them more suitable for risk-averse investors. Cluster 1 represents the moderate risk group, where stocks have medium levels of beta and volatility. These stocks offer a balance between risk and return, meaning they have some growth potential while still maintaining a level of stability. Cluster 2 represents the high risk group, where stocks have higher beta values and higher volatility. These stocks are more sensitive to market movements and experience larger price fluctuations. Although they carry more risk, they also have the potential for higher returns. Overall, the clustering helps to clearly separate stocks based on their risk and return characteristics, making it easier to understand different investment behaviours.

Star Schema Architecture

A data mart for S&P 500 analysis should adopt a star schema to support performance tracking and segmentation.

Dimensions:

- **Time:** Date, month, quarter, and year for longitudinal trend analysis.
- **Company:** GICS Sector, industry classification, and market capitalization size.
- **Geography:** Headquarters location (Country/Region) to identify geographic market trends.
- **Investor Segment:** Retail vs. institutional clusters, identified via the Python-derived risk-return profiles.
- **Market Conditions:** Macroeconomic indicators such as interest rates and VIX (Volatility Index) levels.

Fact Table:

- **Price Dynamics:** Daily Open, Close, High, and Low prices to capture intraday and inter day trends.
- **Liquidity Metrics:** Trading volume and liquidity ratios, indicating the ease of entry and exit for specific stocks.
- **Profitability Ratios:** Daily, monthly, and annual returns representing the growth on investment.
- **Risk Indicators:** Calculated Beta and Annual Volatility values to quantify market sensitivity.
- **Fundamental Values:** Earnings per share (EPS) and P/E ratios to support valuation analysis.

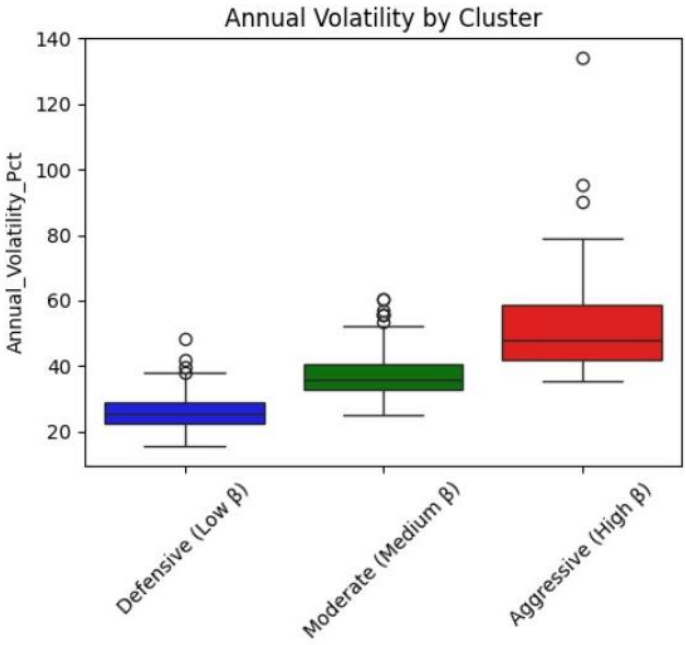
Justification:

These dimensions and metrics enable investors to identify high performing sectors, assess systematic risk, and target specific investor segments. By integrating clustering results directly into the Investor Segment dimension, the business can differentiate between premium investment opportunities and institutional grade liquidity regions, significantly improving strategic portfolio optimisation.

Portfolio Diversification

The business value of our clustering analysis lies in its ability to identify distinct risk reward probabilities, enabling investors to build a diversified portfolio that mitigates unsystematic risk. Our results demonstrate a clear dependency between Beta, Annual Volatility, and Daily Return.

Cluster 2 (Aggressive) represents high beta, high volatility assets that provide the primary engine for portfolio growth during bullish market cycles. However, the high market sensitivity of this cluster necessitates a counterbalance. By integrating Cluster 0 (Defensive) stocks, which exhibit lower volatility and a Beta significantly below 1, investors can create a buffer that protects capital during market downturns. The dependency is such that while Cluster 2 drives Alpha**,** Cluster 0 and the Moderate Cluster 1 ensure portfolio stability. Utilising these Python derived segments allows for strategic asset allocation across the entire S&P 500, ensuring that an investor is not over exposed to any single risk profile, thereby optimising the Sharpe ratio and achieving a more resilient investment structure.



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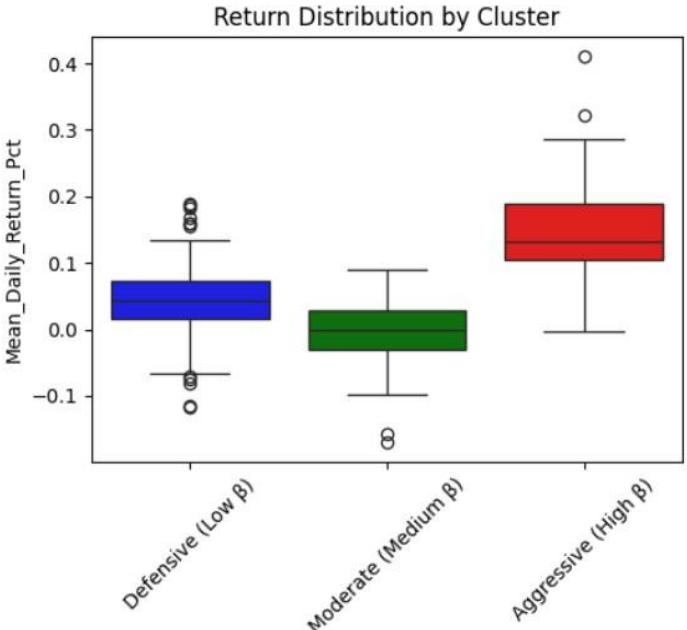

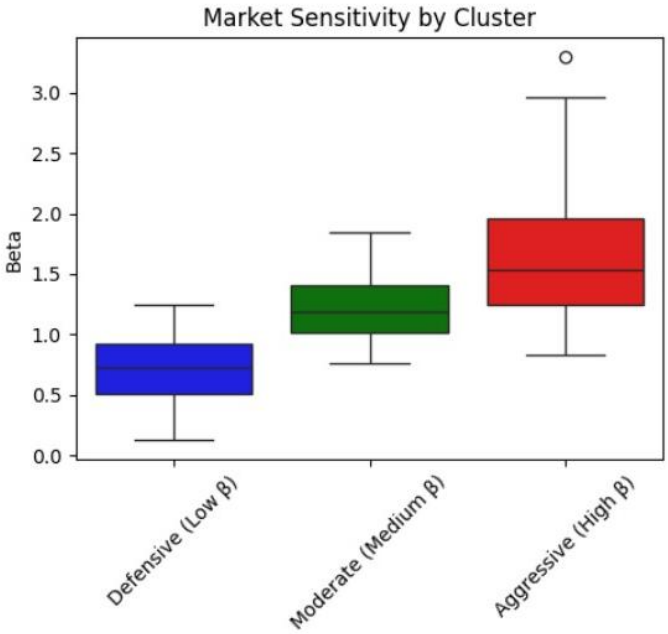

```
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```

```
Sample stocks per cluster:
  Moderate (Medium β): ['A', 'ABNB', 'ACN', 'ADBE', 'ADI']
  Defensive (Low β): ['AAPL', 'ABBV', 'ABT', 'ACGL', 'ADM']
  Aggressive (High β): ['AMD', 'ANET', 'APO', 'APP', 'ARES']
```

The clustering analysis successfully categorised the 496 S&P 500 components into three actionable tiers. Most of the market (321 stocks) falls into the Defensive (Low Beta) category, providing a stable foundation for capital preservation; this includes industry giants like Apple (AAPL) and healthcare leaders such as Abbott Laboratories (ABT) and AbbVie (ABBV). The Moderate (Medium Beta) segment comprises 124 stocks, capturing balanced growth through tech and service firms like Adobe (ADBE), Airbnb (ABNB), and Accenture (ACN). Finally, a specialized group of 51 stocks was identified as Aggressive (High Beta). These high volatility assets, including semiconductor powerhouse AMD and software platform AppLovin (APP), offer the highest potential for Alpha generation. By selecting representative tickers from each of these statistically distinct groups, investors can construct a diversified portfolio that aligns with their specific risk tolerance while maintaining broad exposure to the S&P 500's growth.

When it comes to building the actual portfolio, I'd suggest the following approach to get the best balance of safety and growth. I'd put the lion's share of the money (70% to 80%) into the Defensive Cluster (Cluster 0). Using stable names like Apple and Abbott Laboratories as your foundation really helps protect the capital if the market takes a dip. From there, I'd layer in a 15% to 20% growth slice using Moderate stocks (Cluster 1) like Adobe to make sure the portfolio keeps pace with the S&P 500. Finally, I'd reserve a small 5% to 10% for Aggressive stocks (Cluster 2) like AMD or AppLovin. These are higher risk, but they act as performance boosters that can really drive gains when the market is trending upward. This mix ensures we aren't just chasing returns but actually managing risk across all three groups we identified.

CRM & Sustainability in Fintech

In today's Fintech landscape, Customer Relationship Management systems have evolved from simple sales tracking tools into strategic platforms for embedding Environmental, Social, and Governance principles at the core of operations (Yuan et al., 2025).

A key contribution to social sustainability is the promotion of responsible investment advice and financial inclusion. Our analysis of 496 S&P 500 stocks (after data cleaning in Task A) using agglomerative hierarchical and K-Means clustering produced three distinct risk profiles based on Mean Daily Return, Beta, and Annual Volatility. Cluster 0 represents low risk defensive stocks (low Beta and low volatility), Cluster 1 (moderate Beta and Volatility) and Cluster 2 contains high Beta, high volatility aggressive assets. By integrating these Python-derived cluster labels into a CRM system, Fintech platforms can deliver personalised recommendations and prevent wrongful selling. For example, clients whose risk tolerance aligns with the defensive profile of Cluster 0 can be automatically protected from exposure to Cluster 2 assets during volatile market periods. This data-driven personalisation reduces the risk of unsuitable advice and supports greater financial inclusion for risk-averse or vulnerable investors (Chen et al., 2025).

From an environmental perspective, CRM systems advance sustainability through dematerialisation and operational efficiency. Cloud based CRM architectures eliminate the need for physical branches and paper intensive workflows, significantly reducing a firm's physical and carbon footprint. However, these benefits are not automatic. As proposed in our star schema data mart (Task C.1), consolidating the metrics calculated in Task A (Daily Return, Beta, and Annual Volatility) into efficient structures helps prevent "data bloat". Applying preprocessing techniques such as StandardScaler, Ward's linkage in agglomerative clustering, and validation via the Elbow Method and Silhouette Score reduces unnecessary computational load for real time analytics, thereby lowering energy consumption in data centres and aligning digital operations with global carbon reduction targets (Rahman et al., 2024).

On the governance aspect, CRM platforms serve as a reliable single source of truth. They automate critical compliance processes including Know-Your-Customer (KYC), Anti-Money Laundering (AML) checks, and audit trails. By maintaining a transparent record of how clustering outputs inform client recommendations, Fintech firms can demonstrate ethical decisions and regulatory compliance, particularly with the EU Sustainable Finance Disclosure Regulation (SFDR) (European Commission, 2025).

Critically, challenges persist. Algorithmic clustering risks embedding bias if models are not regularly audited, and the energy demands of large-scale cloud, and AI systems can partially offset environmental gains. Nevertheless, when combined with robust governance frameworks and continuous monitoring, CRM tools transform Fintech platforms from purely transactional services into enablers of sustainable finance.

Ultimately, the clustering analysis conducted in this project supplies the foundational intelligence that CRM systems require to deliver personalised, risk appropriate, and ESG aligned services. By bridging Business Intelligence outputs with customer facing platforms, CRM becomes a powerful driver for delivering economic value, social protection, and environmental responsibility.

References

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